

EDUARDO VASQUEZ

Virtual Production Technical Artist | Researcher | Visual Artist

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 Hamilton, ON

Professional Summary

Versatile Technical Artist and Software Developer with 5+ years creating interactive digital experiences across education, broadcasting, and experimental media. Expert in real-time visual systems using Unreal Engine, React Three Fiber, and custom web technologies, with a proven track record of optimizing performance while maintaining creative vision. Successfully delivered 20+ features ranging from VR applications to production websites, consistently collaborating across multidisciplinary teams. Passionate about emerging technologies and committed to building accessible, human-centered digital experiences.

Technical Skills

Software & Tools

- Real-time Engines: Unreal Engine 4/5, Reality Editor (Zero Density), TouchDesigner
- 3D/Animation: Blender, Maya, Houdini
- Programming/Scripting: JavaScript, Python, C++, HLSL, GLSL, WebGL
- Web Development: JSX, React, R3F, Three.js, A-Frame, jQuery, HTML5, CSS3
- Creative Development: WebXR/VR, Canvas 2D, Interactive Web Experiences
- Virtual Production: XR Broadcasting, Real-time Compositing, Motion capture, Tracking systems and Physical Computing integration, Interactive Set Design, Custom Shaders
- Adobe Creative Suite: Photoshop, After Effects, Premiere Pro
- Additional: Substance Suite, Live Streaming Platforms

Technical Specializations

- Real-time Visual Systems
- XR/VR/AR Development
- Shader Development & Materials

- Interactive Media Design
 - Virtual Production Pipelines
 - Data Visualization
 - Procedural Graphics Generation
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Professional Experience

Technical Artist | LUV (Laboratoire d'Univers Virtuels) | TFO

2021-2025 | Toronto, ON

- Developed interactive XR broadcasting solutions using Unreal Engine and Reality Engine, enhancing live production capabilities for educational media content at the LUV for over 20 features including full series, mini series, capsules, one VR experience and a video-game prototype.
- Designed and implemented interactive solutions through the UE blueprint system and C++
- Designed and implemented custom interactive tools for the operators and switchers in studio
- Collaborated with production teams to design and implement real-time visual effects, particle systems in UE Niagara and shaders for live broadcast
- Optimized rendering pipelines for broadcast-quality output while maintaining real-time performance requirements over 60 fps.
- Developed VR applications for promotion and marketing events.
- Accompanied the implementation and troubleshooting of the LUV motion capture system Blacktrax.

Founder / Developer | NIKAI INTERACTIVE

2020-Now | Toronto, ON

- Founded interactive digital art brand and developed portfolio website using JavaScript, Three.js, and jQuery.
- Led development of Le Labo Art website (lelaboart.github.io) using React and Create React App, guided the organization through their database establishing process on Airtable.
- Built experimental React applications including 'tweeterio' (Twitter clone) and speakio, demonstrating full-stack development capabilities
- Developed 'rect-three-fiber_test' project exploring React Three Fiber for immersive web experiences
- Designed and implemented immersive web experiences showcasing interactive media capabilities

- Created custom web-based applications for digital art installations and client projects
- Managed full-stack development process from concept to deployment for interactive media projects

Technical Director / Visual Designer | TALK MEDIA

2020 | Toronto, ON

- Led technical and visual production for "Shiny Talking People" and "Time Machine: Live Online" online events
- Managed live streaming infrastructure and resolved real-time technical issues during production
- Developed custom real-time visual assets for live broadcast enhancement
- Coordinated production design workflows integrating traditional media with interactive elements

Technical Operator / Streamer | CARIBBEAN TALES FILM FESTIVAL

2018 | Toronto, ON

- Managed multi-platform streaming systems for live festival events and interviews
- Provided technical support and troubleshooting for live broadcast operations
- Ensured seamless streaming delivery across multiple platforms during high-profile events

Education

Master of Arts in Communication - Experimental and New Media

Université du Québec à Montréal (UQAM) 2012-2016

- Specialization in interactive media and digital communication technologies

Bachelor of Arts and Science in Animation

Université Laval 2009-2012

- Focus on digital animation and visual effects production

Software Development Bootcamp

Lighthouse Labs 2020

- Full-stack web development with emphasis on JavaScript and modern frameworks

Direction of Photography

Ecole de Cinéma d'été de Percé 2025

- Intensive cinematography training with Yves Bélanger, ASC, CSC
 - Visual composition, lighting design, and cinematic camera work
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Teaching & Research Experience

CG Instructor | LE LABO Toronto

2021 | Toronto, ON

- Delivered technical training in Blender and TouchDesigner.
- Designed and built a medium scale paper sculpture for projection mapping
- Developed curriculum for video mapping techniques, real-time graphics and procedural animation workflows that led to the creation of two augmented sculpture features presented at Le Labo

Visual Arts Instructor | Alliance Française de Toronto

2017 – 2019 | Toronto, ON

- Taught traditional media production and visual storytelling techniques
- Mentored students in animation, photography and experimental media development

Documentary Filmmaking Facilitator | Oxfam Quebec / Wapikoni Mobile

2015

- Facilitated community-based documentary production workshops that led to the production of 6 short films
- Provided technical training for indigenous filmmakers and media creators from two bolivian first nation, the Yampara people and the Qhara-Qhara people

Research Agent | Communautique / Echofab

2014

- Prototyped an immersive 360-degree video device

- Conducted R&D on emerging media and rapid prototyping technologies for community applications
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Additional Information

Languages: English, French, Spanish (Professional proficiency)

Professional Recognition:

- Jury Member, Canada Arts Council "Explore and Create" Program (2021)
- Speaker, AGAFV Forum Ottawa (2019)
- Speaker, Université de Montréal "Authentic Artifice" Colloquium